Aphasias

Reminder: Reasoning about aphasia Before Chomsky

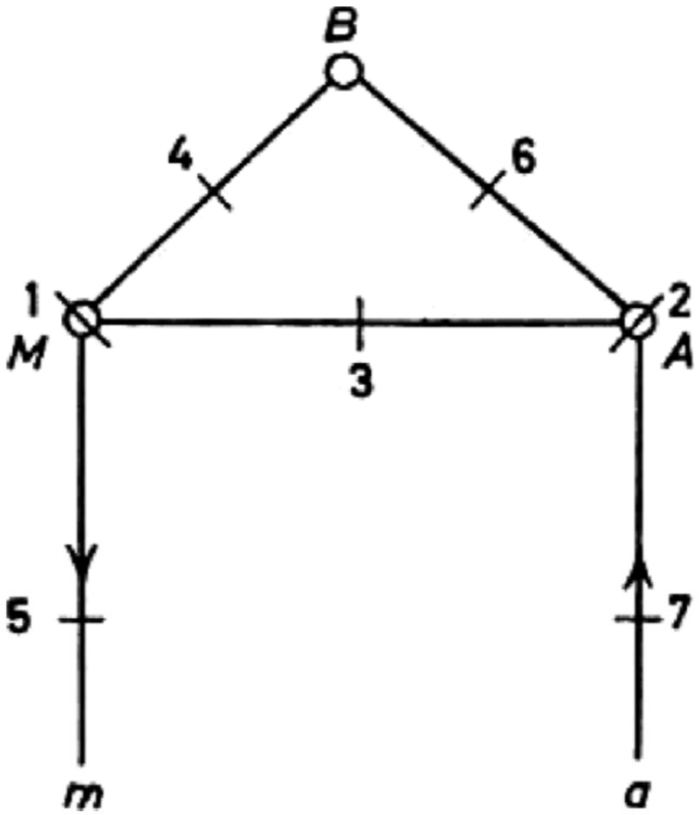


Fig. 36.5. Lichtheim's "House" model. A represents the center for sound images; M is the center for movement images; B is the center for concepts; a is the acoustic pathway and m is the motor pathway. 1–7 represent various forms of aphasia, depending on the destruction of a center or pathway.

Eling, P., & Whitaker, H. (2009). History of aphasia: from brain to language. In *Handbook of clinical neurology* (Vol. 95, pp. 571-582).

In the 1870's a number of German neurologists — Lichtheim, Wernicke — began making more systematic observations of aphasia.

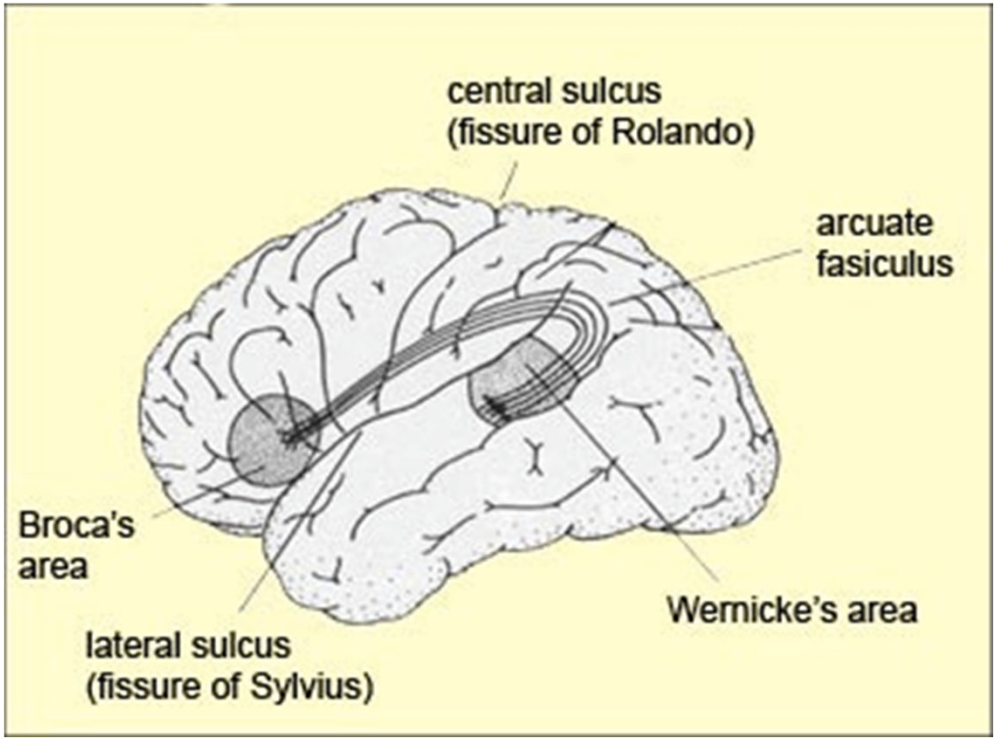
They ascribed specific losses of function to "disconnections" between "centers."

So, a loss of the ability to understand spoken words might be due to a disconnection between "A" and "B."

A loss of the ability to do verbatim repetition may be due to a disconnection between "A" and "M," and so on.

This kind of "box and arrow" model is foundational, not just in aphasiology, but in modeling of typical language function (consider the interactive activation model we discussed in the last lecture).

The “Geschwind-Wernicke model” of brain and language



In the 1970’s Norman Geschwind revived this approach.

In the century between Wernicke and Geschwind, the field of linguistics evolved dramatically.

Specifically, the development of a theoretical model in which *syntax* was independent of other aspects of language, had its own computational principles, etc., opened the door for theorizing about the (*double) dissociations* of language functions in more sophisticated ways.

ELSEVIER

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Lingua

Some thoughts about the brain/mind/language interface☆

Victoria A. Fromkin

The speech output of Broca's aphasia patients is characterized by word-finding pauses, loss of grammatical morphemes, and quite often, disturbed word order. Auditory comprehension for colloquial conversation gives the impression of being generally good, although controlled testing reveals considerable impairments. The term agrammatism is almost synonymous with Broca's aphasia, although some patients with lesions in Broca's area are not agrammatic and some agrammatics would not classify neurologically as Broca's aphasics.

Wernicke's aphasia patients, on the other hand, produce fluent speech with good intonation and pronunciation, but with many word substitutions (both semantically similar and dissimilar), neologisms as well as phonological errors. They also show comprehension difficulties.

An attractive dichotomy:

Wernicke's aphasia = "Word salad" or "Jabberwocky," often with neologisms. Syntax roughly intact:

> One Wernicke's aphasia patient, for example, replied to a question about his health with: "I felt worse because I can no longer keep in mind from the mind of the minds to keep me from mind and up to the ear which can be to find among ourselves" (Kriendler et al., 1971). Another patient, when asked about his poor vision said: "My wires don't hire right". And an aphasic physician, asked if he was a doctor, replied: "Me? Yes, sir, I'm, a male demaploze on my own. I still know my tubaboys what for I have that's gone hell and some of them go". One aphasic described a fork as "a need for a schedule" and a spoon as "How many schemes on your throat".

Broca's aphasia = "Telegraphic" speech, difficulty with word order and grammatical particles (relative sparing of important content words):

> These fluent but uninterpretable utterances are very different from the Broca aphasic's answer when asked what brought him to the hospital: "Yes – ah – Monday – ah – Dad and P.H. (the patient's name) and Dad ... hospital. Two ... ah doctors ..., and ah ... thirty minutes ... and yes ... ah ... hospital. And, er Wednesday ... nine o'clock and eh Thursday, ten o'clock ... doctors. Two doctors ... and ah ... teeth. Yeah ... fine" (Badecker and Caramazza, 1985).

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Note that this is fairly distant from how Broca described his original cases.

Remember this slide from a few weeks ago?

This is confusing: *Broca's actual patients would not necessarily be classified as Broca's aphasics.* They were arguably unable to produce speech at all, apparently due to motor control issues, and would be classified as suffering from *apraxia of speech.*

Paul Broca's presentation of Leborgne (or "Tan")



Behavioral description:

- could only produce one jargon syllable
- seemed otherwise to be in possession of his senses
- in fact, seemed to *understand what was being said.*

Consider:

- the role of dissociations in this argument: why is it critical that Leborgne has other senses intact?

- why does a "capacity for articulate speech" sound like a more plausible function of a particular region of the brain than "criminality" or "philopropogenitiveness?"

- if you could examine Tan, what measurements would you want to make in order to specify his deficits in terms of the levels of description used in modern linguistics (e.g., phonetic, phonological, lexical, syntactic)?

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

This dichotomy is interesting in part because it seems well aligned with a sharp distinction drawn late 20th Century formal linguistics between the *grammar* and the *lexicon*.

This starts with Chomsky's formulation of the computational problem of language learning, and compelling demonstrations of the independence of syntax and semantics.

It appears that we recognize a new item as a sentence not because it matches some familiar item in any simple way, but because it is generated by the grammar that each individual has somehow and in some form internalized. And we understand a new sentence, in part, because we are somehow capable of determining the process by which this sentence is derived in this grammar.

"Neither (a) 'colorless green ideas sleep furiously' nor (b) 'furiously sleep ideas green colorless', nor any of their parts, has ever occurred in the past linguistic experience of an English speaker. But (a) is grammatical, while (b) is not." (Chomsky, 1955)

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Consider, too, the difference between the Functional and Positional levels in our speech production model.

Lemmas have an *arbitrary* relationship to their referents.

But the **syntactic trees**, **affixes**, etc. at the positional level are the product of *abstract syntactic processes.*

"But," you might object "what about words like HOUSEBOAT and DOGHOUSE, where the meaning is clearly not arbitrary," and I would remind you of speech error evidence that *derivational morphology* is done at the positional level, e.g., *easy enoughly.*

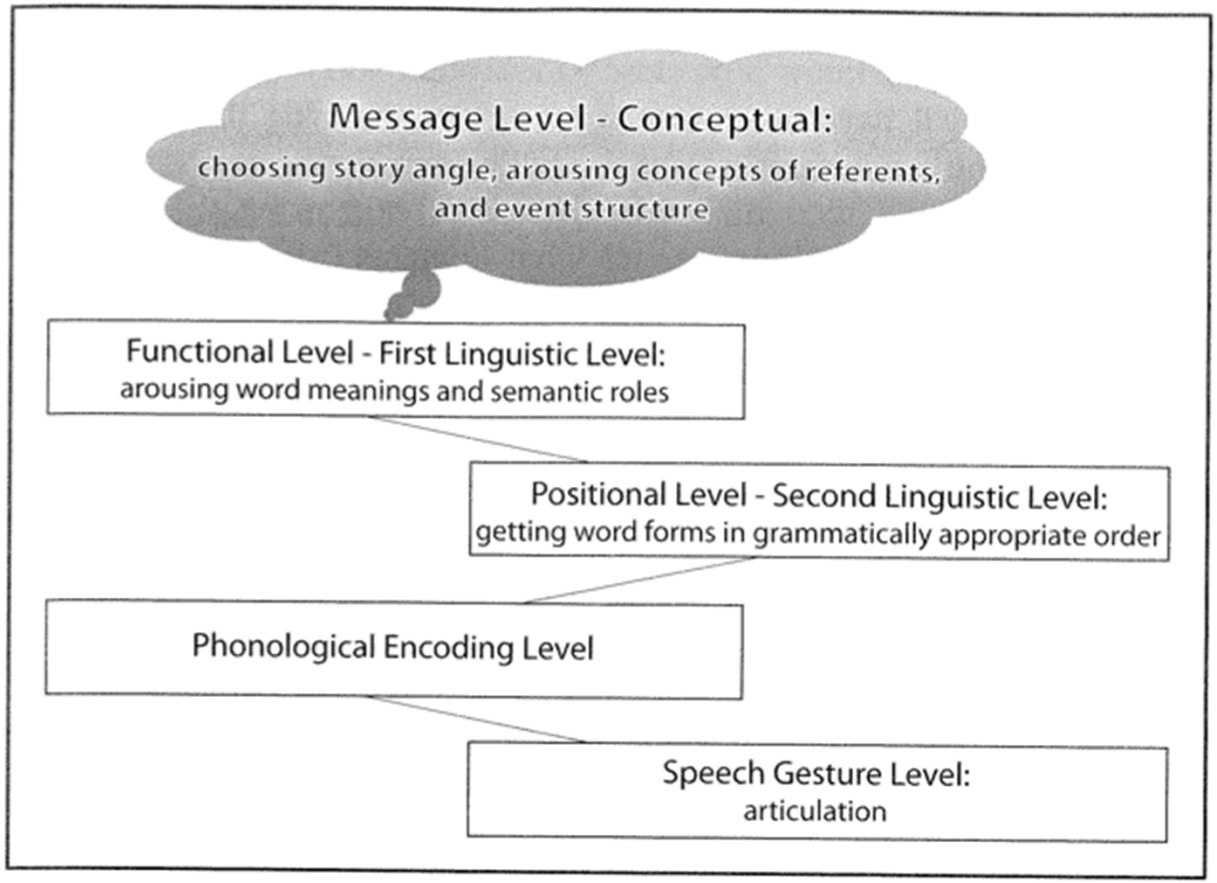


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Remember, also, that there are popular theories of *sentence processing* in which the syntactic structure of sentences is computed independently of meaning as we listen/read.

Isolating these "parsing" mechanisms requires carefully constructing stimulus materials.

One possibility then, is that Broca's patients *generally understand what's being said* even though they have a *core grammatical deficit.*

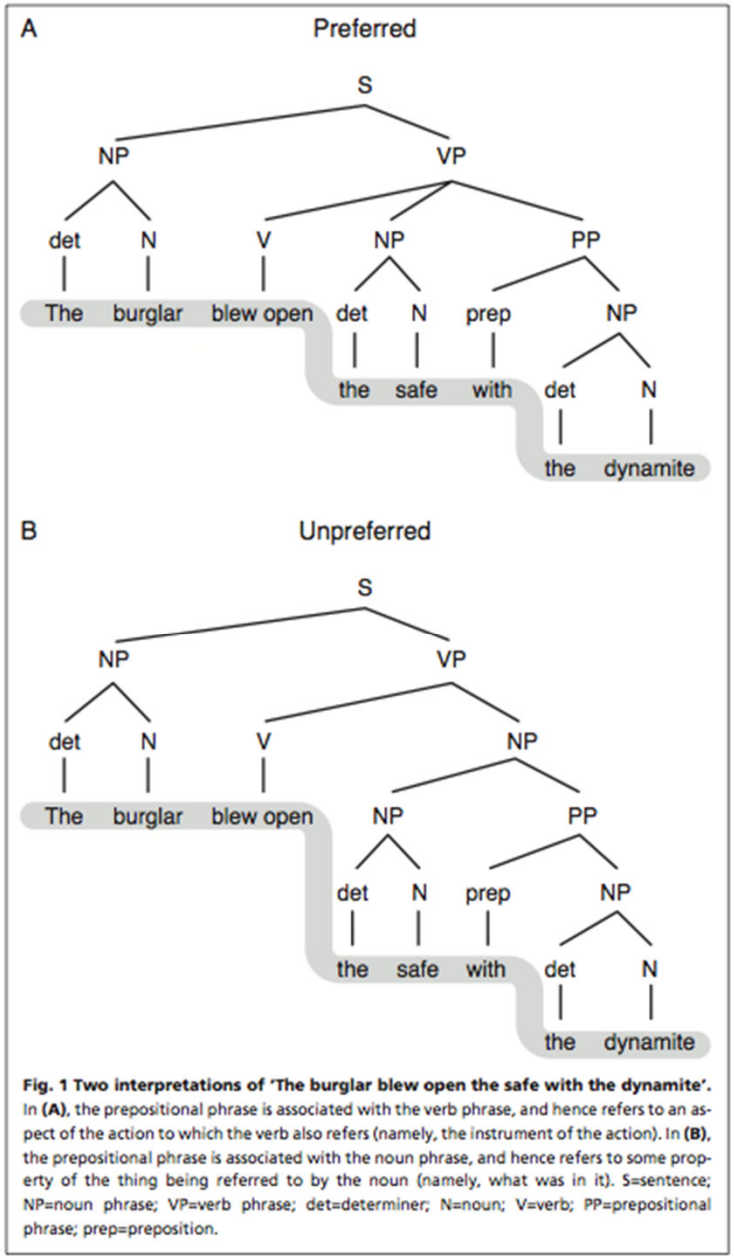


Fig. 1 Two interpretations of 'The burglar blew open the safe with the dynamite'. In **(A)**, the prepositional phrase is associated with the verb phrase, and hence refers to an aspect of the action to which the verb also refers (namely, the instrument of the action). In **(B)**, the prepositional phrase is associated with the noun phrase, and hence refers to some property of the thing being referred to by the noun (namely, what was in it). S=sentence; NP=noun phrase; VP=verb phrase; det=determiner; N=noun; V=verb; PP=prepositional phrase; prep=preposition.

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Remember that none of this theory was available to Broca back in the 1860's.

For him, it was interesting enough to dissociate "articulate speech" from comprehension.

But say we wanted to demonstrate that there are patients whose difficulty is more generally with *syntax.*

How would we do it?

Paul Broca's presentation of Leborgne (or "Tan")



Behavioral description:

- could only produce one jargon syllable
- seemed otherwise to be in possession of his senses
- in fact, seemed to *understand what was being said.*

Consider:

- the role of dissociations in this argument: why is it critical that Leborgne has other senses intact?

- why does a "capacity for articulate speech" sound like a more plausible function of a particular region of the brain than "criminality" or "philopropogenitiveness?"

- if you could examine Tan, what measurements would you want to make in order to specify his deficits in terms of the levels of description used in modern linguistics (e.g., phonetic, phonological, lexical, syntactic)?

Why was it interesting to show that Broca's patients have comprehension deficits?

Broca's aphasia is almost always accompanied with some level of motor dysfunction.

If agrammatic speech merely reflects an attempt to communicate despite motor limitations.

Remember Hughlings Jackson: We are observing what the individual can do despite the impairment. Just because a brain injury leads to agrammatic speech doesn't literally mean that the grammar itself is injured!

Consider also Chomsky's distinction between *competence* and *performance*. When motor deficits obviously interfere with performance, comprehension may provide a clearer window into competence…

& Kellar, 1973; Lenneberg, 1973). Indeed, accepting that anterior damage spares the patient's tacit knowledge of his language, one current version (Lenneberg, 1973) attributes the telegrammatic output to a neuromuscular problem: The strain of speaking is so great that the patient speaks asyntactically in order to economize effort.

Caramazza & Zurif (1976)

Man Bites Dog

Why do we need psycholinguistic tests to observe grammatical difficulties in comprehension?

In conversation, there are lots of redundant cues to what a speaker intends:

The dog bit the man.

There are so many ways to get the right meaning here:

- Most likely combination of DOG, BITE and MAN into an event representation with or without syntax.

- Canonical word order: Agent – Verb – Undergoer; Subject – Verb – Object

Further, a lot of language comprehension takes place in discourse context that clarifies a lot...

Maybe Broca's patients' comprehension depends on these compensatory *heuristics*. If that were the case you might want to test them with sentences more like:

The dog that the man is biting is angry.

Indeed, Caramazza and Zurif devised a list of just this kind of sentence, and a set of control sentences.

TABLE 1

SENTENCE MATERIAL USED IN THE EXPERIMENT

A. Control sentences
1. The girl is kicking a green ball.
2. The cat is chasing a blue bird.
3. The girl is pulling a red wagon.
4. The man is reading a yellow book.
5. The boy is riding a new bicycle.
6. The girl is holding a broken doll.
7. The boy is throwing a round ball.
8. The boy is carrying a sharp pencil.

B. Center-embedded sentences with semantic constraints
9. The apple that the boy is eating is red.
10. The wagon that the horse is pulling is green.
11. The book that the girl is reading is yellow.
12. The bat that the boy is carrying is brown.
13. The bicycle that the boy is holding is broken.
14. The coat that the girl is wearing is new.
15. The dress that the woman is washing is torn.
16. The balloon that the clown is holding is round.

C. Reversible center-embedded sentences
17. The cow that the monkey is scaring is yellow.
18. The horse that the bear is kicking is brown.
19. The cat that the dog is biting is black.
20. The fish that the frog is biting is green.
21. The lion that the tiger is chasing is fat.
22. The girl that the boy is hitting is tall.
23. The man that the woman is hugging is happy.
24. The girl that the boy is tickling is happy.

D. Improbable center-embedded sentences
25. The horse that the girl is kicking is brown.
26. The lion that the baby is scaring is yellow.
27. The bird that the worm is eating is blue.
28. The dog that the man is biting is black.
29. The man that the horse is riding is fat.
30. The woman that the boy is spanking is sad.
31. The policeman that the robber is chasing is skinny.
32. The girl that the cat is patting is small.

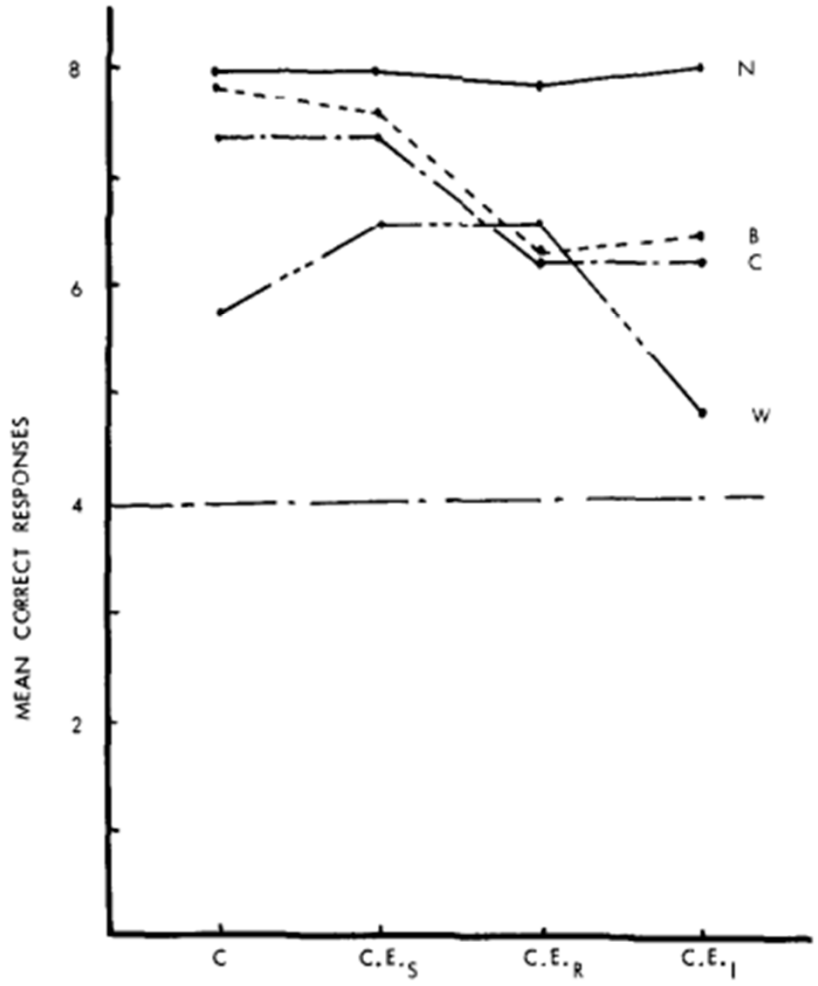


FIG. 1. Mean correct performance for each patient group (N = Normal, B = Broca, C = Conduction, and W = Wernicke) as a function of sentence type (CE_S = semantically constrained, CE_R = reversible, CE_I = improbable, and C = control).

Data from a sentence-picture matching task.

Healthy controls are more or less equally good at all stimulus types.

Wernicke's patients are poor overall. This is probably a result of their comprehension deficits.

The Broca's patients seem to have a particular difficulty keeping "who did what to whom" straight when syntax conflicted with common sense, probable semantic roles for nouns, etc.

This *specific* comprehension deficit was taken as evidence that Broca's aphasia impacts *syntax* specifically.

So we can see how a more specific theory of what the “parts of language” might be can open up new ways of exploring structure-function relationships.

Rather than just breaking up aphasics into those with good and poor comprehension, or those who are more or less fluent, we can isolate problems with particular aspects of language, i.e.

The Grammar vs. The Lexicon

Problems with this dichotomy

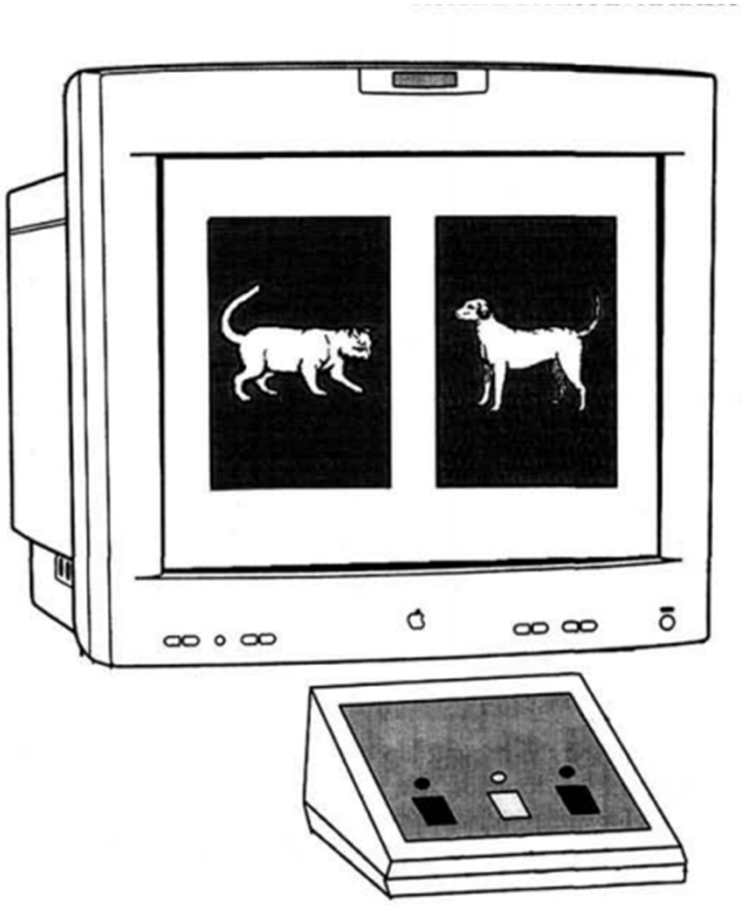


Figure 4. Experimental setup and sample visual stimuli (Experiments 1A and 1B).

Dick et al. had participants listen to four different types of sentences:

Active:

The dog is biting the cat.

Subject cleft:

It's the dog that is biting the cat.

Object cleft:

It's the cat that the dog is biting.

Passive:

The cat is bitten by the dog.

Instructions: Indicate which animal is *doing the action.*

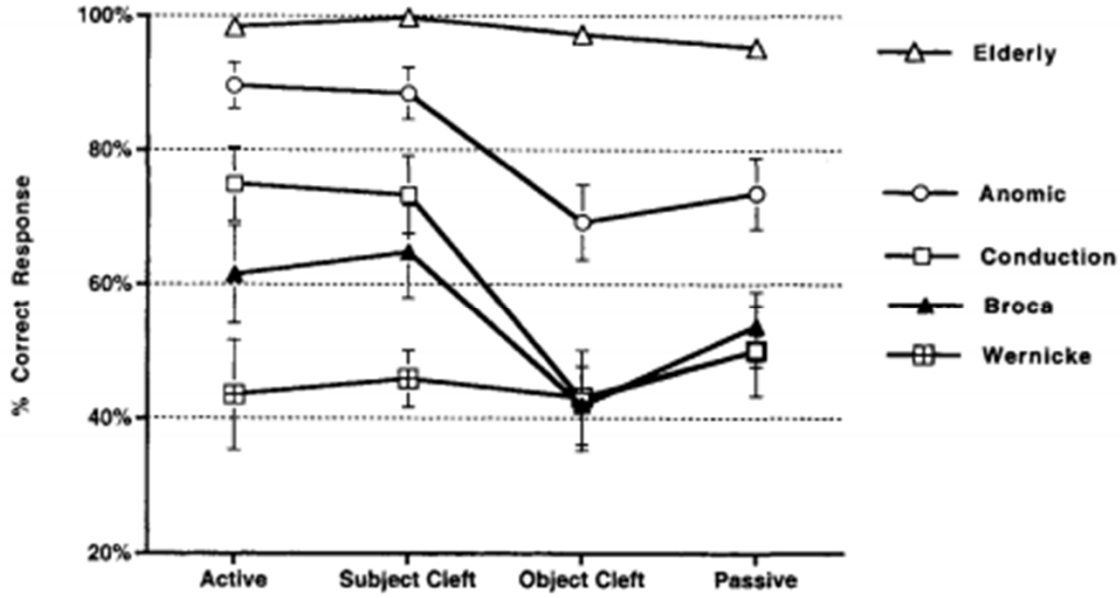


Figure 5. Percentage of correct responses by sentence type for aphasic patients (grouped by the Western Aphasia Battery) and older control participants (Experiment 1A). Error bars represent ±1 *SEM*.

When aphasia patients and healthy aging controls do this task, a pattern emerges much like what Caramazza and Zurif observed. The Broca's aphasia patients have particular difficulty with the syntactically complex stimuli (although note that their performance is overall not so great).

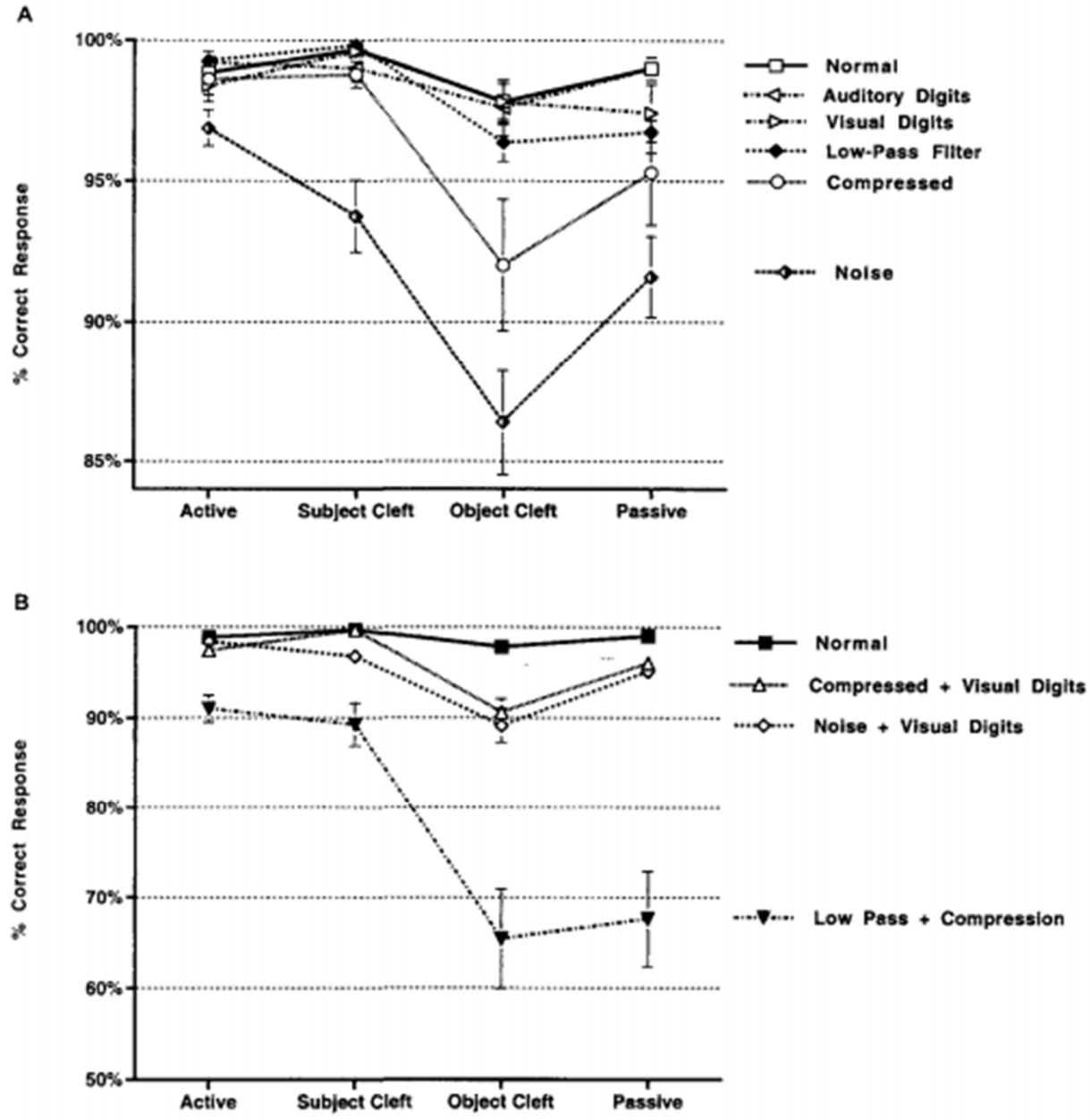


Figure 9. A: Percentage of correct responses by sentence type for students under single-stress conditions (Experiment 1B). B: Percentage of correct responses by sentence type for students under dual-stress conditions (Experiment 1B). Error bars represent ±1 *SEM*.

These are data from healthy undergraduates under conditions that stress the language system in different ways.

Note that the object cleft and passive sentences are particularly vulnerable to all manipulations.

The “stress” conditions were things like speeding up the speech, or asking people to remember lists of numbers as a secondary task.

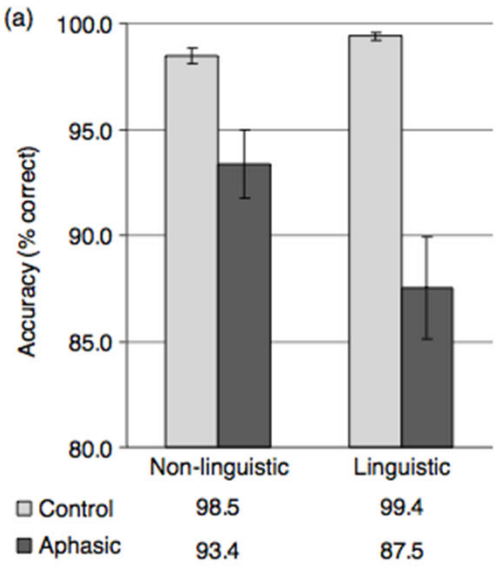
This suggests that the comprehension deficits seen in Broca’s aphasia patients may be a “performance” issue after all.

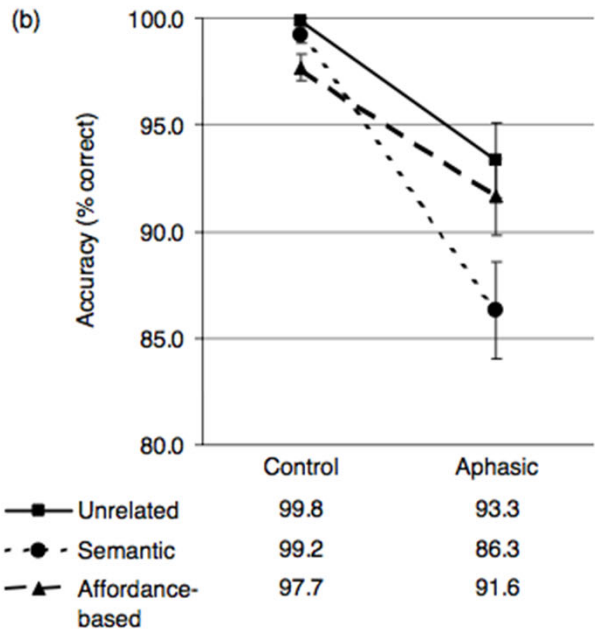
Are Broca's aphasics comprehension deficits language-specific?



Saygin et al. (2004) asked patients to match the action to the object.

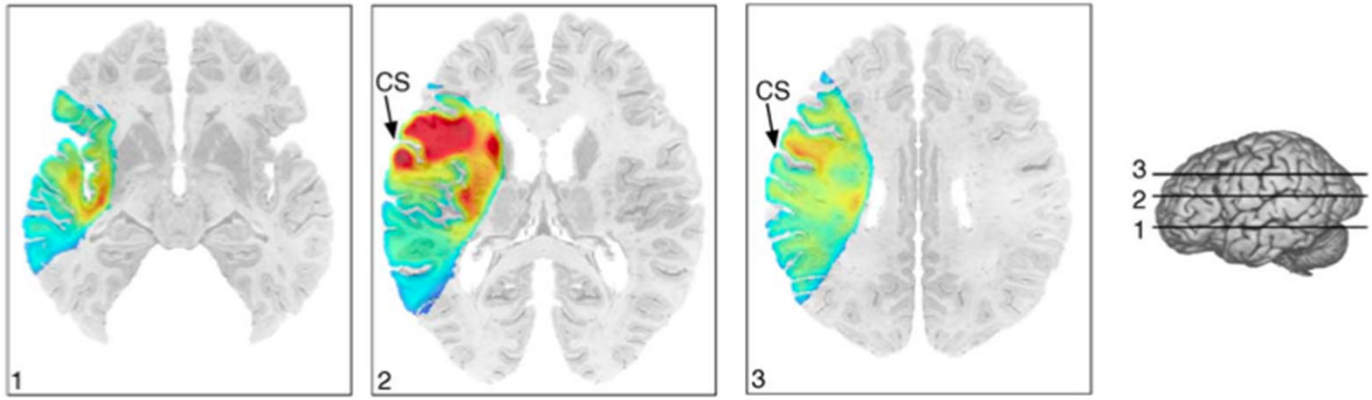
Broca's aphasics had difficulty with this task, whether the stimuli were linguistic or not.





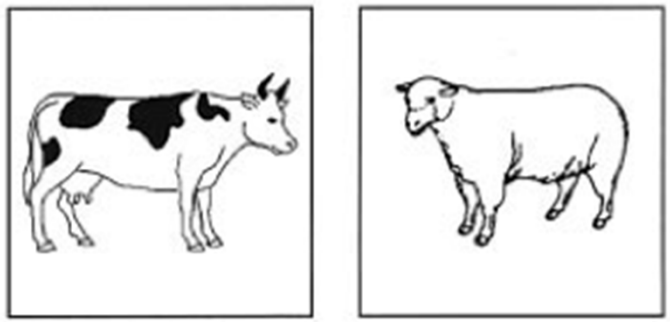
Are Broca's aphasics comprehension deficits language-specific?

(a) Lesion correlates of non-linguistic (pantomime interpretation)deficits

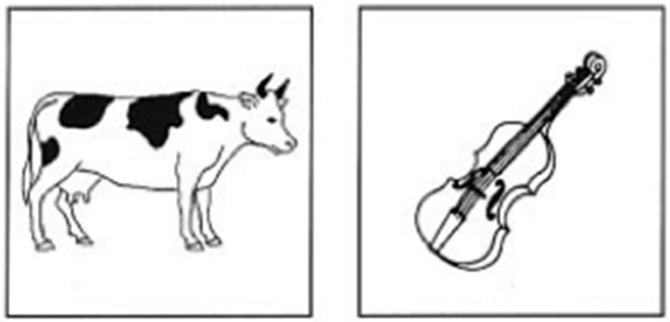


Further, performance in the non-verbal version of this task was significantly correlated with damage to Broca's Area.

Are Wernicke's aphasics comprehension deficits language-specific?



= Mooooooo...



= "cow mooing"

Saygin et al. asked patients to match environmental sounds to pictures.

Wernicke's patients were particularly bad at this task.

No significant difference was observed between verbal and non-verbal versions of the task.

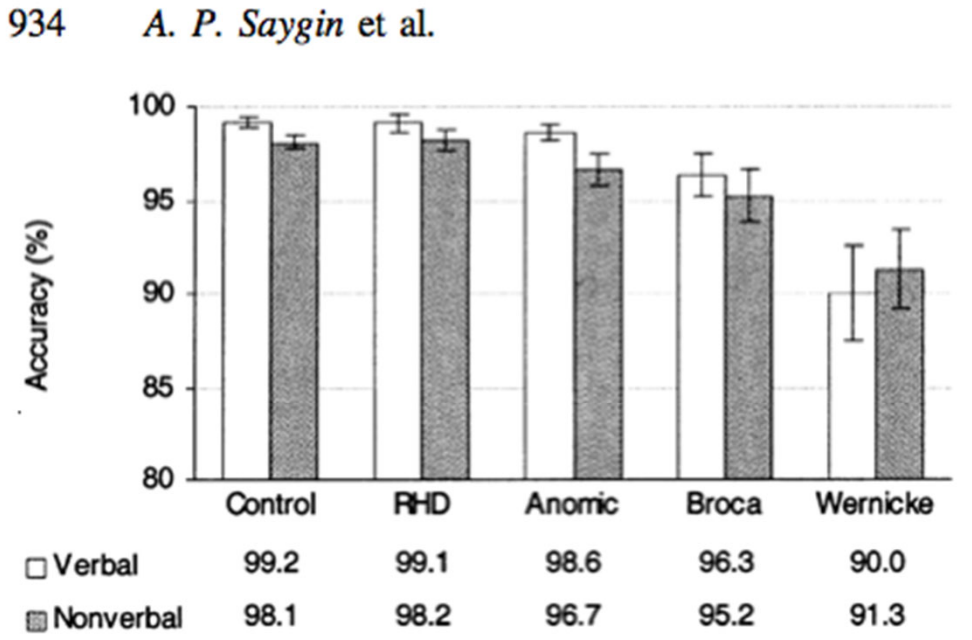
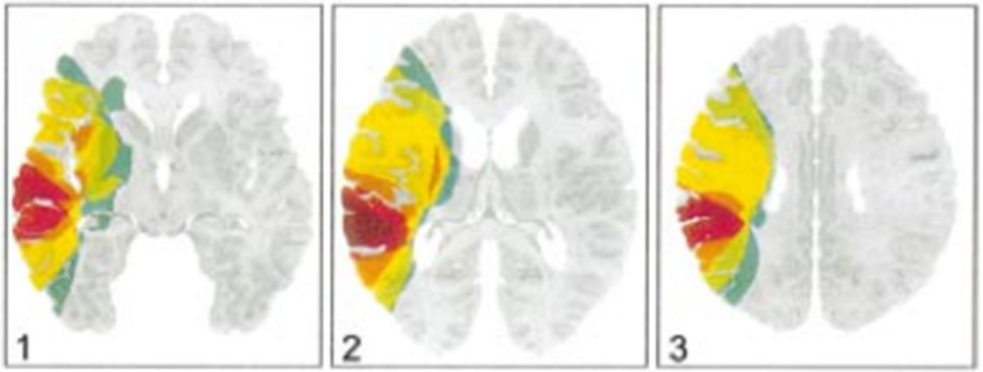


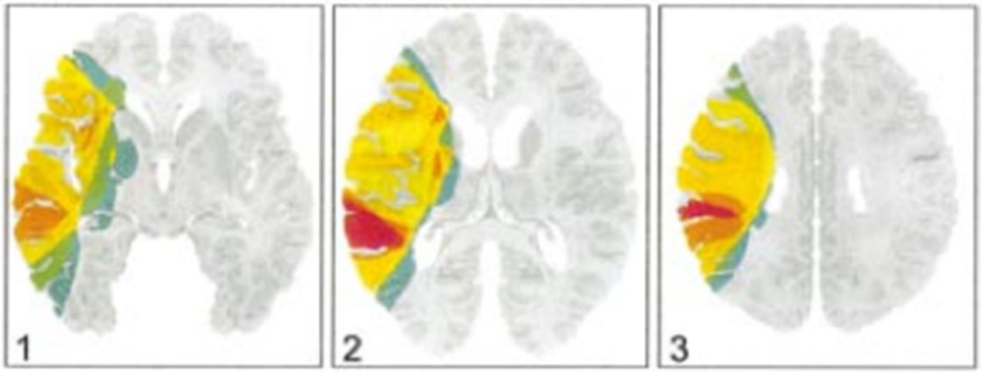
Fig. 2 Accuracy depicted across verbal and nonverbal domains for all subject groups. Groups differed in their overall accuracy ($P < 0.0001$) with control = RHD = anomic > Broca's > Wernicke's (all comparisons corrected $P < 0.01$).

Lesion overlays based on accuracy

Impaired in NONVERBAL (n = 8)



Impaired in VERBAL (n = 10)



Deficits in both verbal and non-verbal sound-picture matching are associated with Wernicke's area.

So, how language-specific are the deficits in aphasia?

This is still a controversy, actually.

What should be clear from this discussion is how the terms of the debate have evolved.

- relatively sophisticated tests to tap specific aspects of language

- quantitative analyses of behavior

- quantitative analyses of structure-function relationships

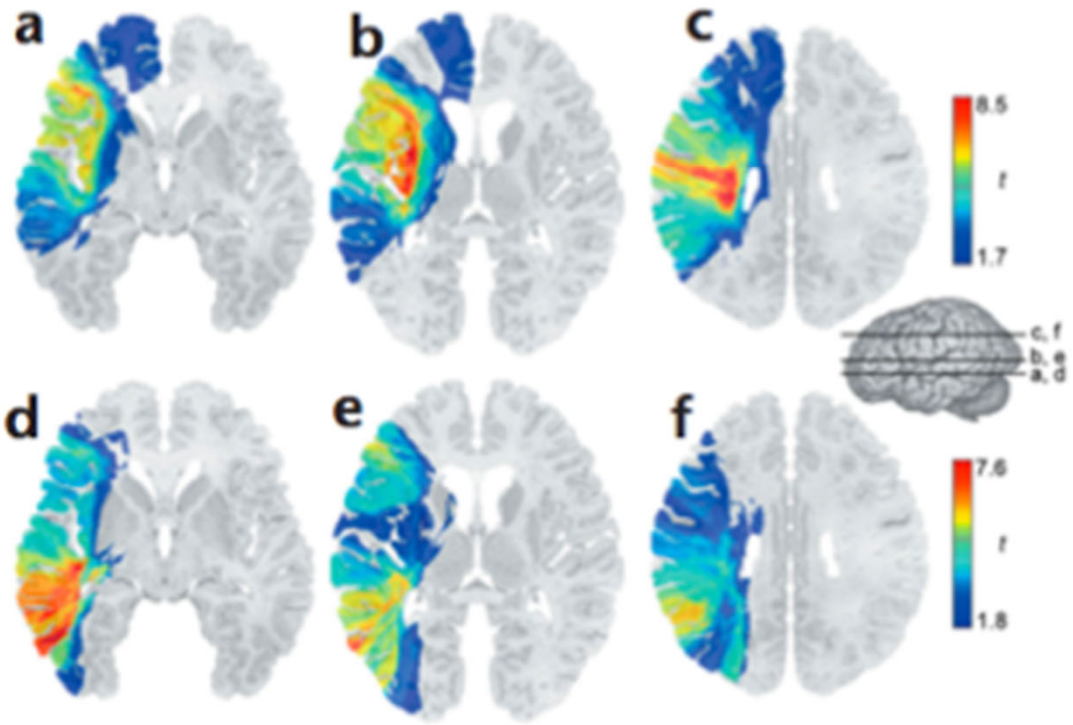
We actually have come quite a long way from the 19th Century!

For Thursday:

Language development…

Voxel-based lesion–symptom mapping

Elizabeth Bates[1], Stephen M. Wilson[2], Ayse Pinar Saygin[1], Frederic Dick[1], Martin I. Sereno[1], Robert T. Knight[3,4] and Nina F. Dronkers[4,5]



Formalizing structure-function relationships in aphasia data (MRI data)

The "symptoms" here are defined very broadly:

(a-c) show regions that are likely to be damaged in individuals with *fluency* deficits.

(d-f) show regions associated with *comprehension* deficits

So, data from a large number of patients is consistent with a rough double dissociation, where posterior regions are associated with perception, whereas anterior regions are associated with production.